EXPERIMENT No 14:
Automatic Dustbin Using Arduino

Student Guide

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Objective:

The goal of this project is to design and implement an automatic dustbin using an Arduino microcontroller and an ultrasonic sensor. This dustbin will automatically open its lid when it detects an object (such as a hand) within a specific range.

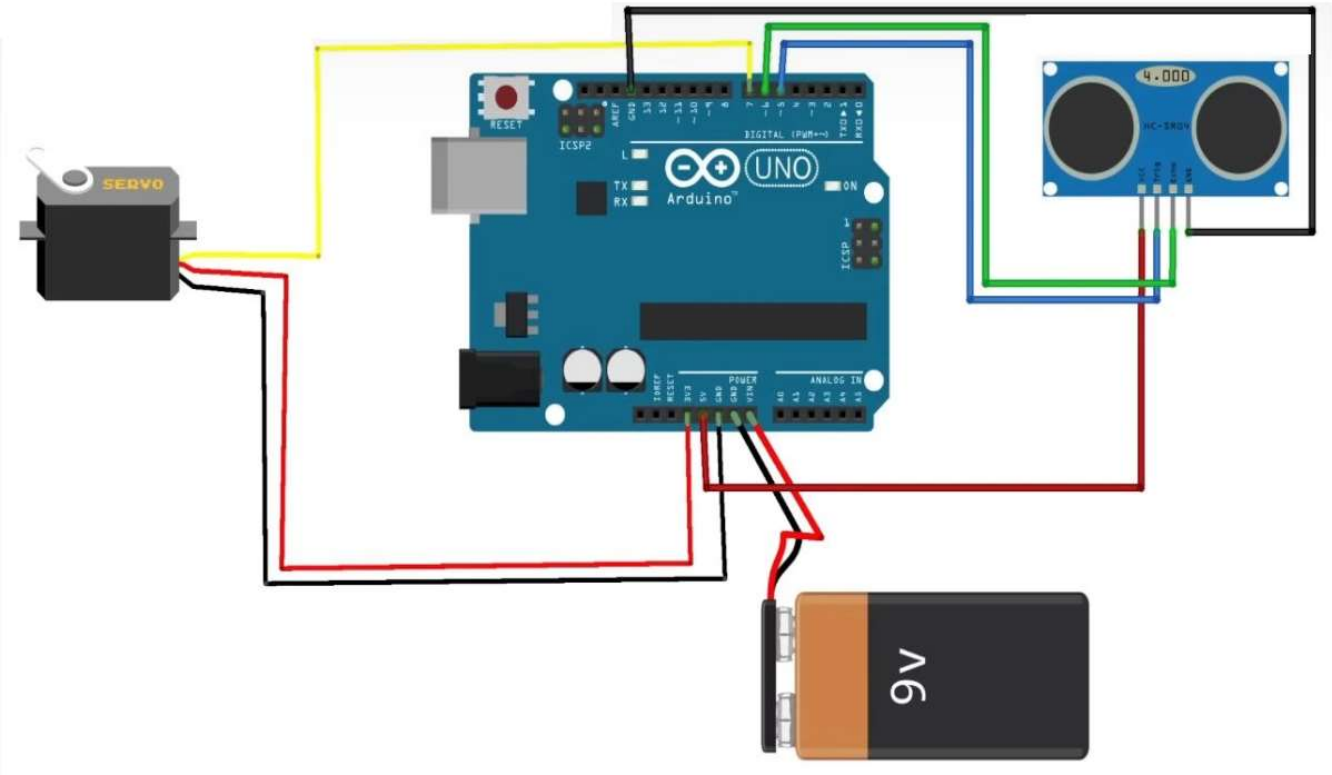
Materials Needed:

1. **Arduino Uno R3** – Microcontroller to control the system.
2. **Ultrasonic Sensor (HC-SR04)** – Used to detect the presence of objects at a certain distance.
3. **Servo Motor** – To open and close the dustbin lid.
4. **Jumper Wires** – For making connections.
5. **Breadboard** – For easy wiring.
6. **Dustbin** – To integrate the system.
7. **Power Supply** – 5V power supply or USB connection.
8. **Resistors (optional)** – Depending on sensor connections.
9. **Buzzer or LED (optional)** – To give a signal when the dustbin is open.

How It Works:

- The ultrasonic sensor measures the distance of an approaching object (like a hand).
- If the object is within a certain range (e.g., less than 20 cm), the Arduino sends a signal to the servo motor.
- The servo motor then rotates to open the dustbin lid.
- After a short delay, the servo motor closes the lid automatically.

Circuit Diagram:



- **Ultrasonic Sensor (HC-SR04)**:
  - VCC → 5V (Arduino)
  - GND → GND (Arduino)
  - TRIG → Digital Pin 9
  - ECHO → Digital Pin 10
- **Servo Motor**:
  - VCC → 5V (Arduino)

- GND → GND (Arduino)
- Control Signal → Digital Pin 6

Code Explanation:

cpp

Copy code

```
#include <Servo.h>        // Include Servo library

Servo myServo;            // Create a Servo object
int trigPin = 9;          // TRIG pin of ultrasonic sensor
int echoPin = 10;         // ECHO pin of ultrasonic sensor
int servoPin = 6;         // Servo control pin
long duration, distance;  // Variables to store distance

void setup() {
  pinMode(trigPin, OUTPUT);  // Set TRIG pin as OUTPUT
  pinMode(echoPin, INPUT);   // Set ECHO pin as INPUT
  myServo.attach(servoPin);  // Attach the servo to pin 6
  myServo.write(0);          // Start with dustbin closed
  Serial.begin(9600);        // Initialize serial communication
}

void loop() {
  // Send a 10us pulse to TRIG to start the sensor
```

```
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Measure the time for ECHO to go HIGH
  duration = pulseIn(echoPin, HIGH);

  // Calculate distance in cm
  distance = duration * 0.034 / 2;

  if (distance < 20 && distance > 0) {  // If an object is detected within 20 cm
    myServo.write(90);  // Open the dustbin (90 degrees)
    delay(3000);        // Keep it open for 3 seconds
    myServo.write(0);   // Close the dustbin (0 degrees)
  }

  delay(500);           // Small delay before the next reading
}
```

Steps to Build:

1. **Connect the Ultrasonic Sensor** to the Arduino as per the circuit diagram.
2. **Connect the Servo Motor** to the appropriate pins on the Arduino.
3. Upload the provided **Arduino code** to the Arduino board.
4. **Power up the system** either using a USB cable or an external 5V power supply.
5. Test the system by placing your hand near the sensor, and watch as the dustbin lid opens and closes automatically.

Troubleshooting:

- **No Movement from Servo Motor**: Ensure the servo motor is correctly connected and powered.
- **Ultrasonic Sensor Not Detecting**: Check the wiring of the TRIG and ECHO pins.
- **Distance Measurements are Off**: Adjust the distance threshold in the code for better responsiveness.

Learning Outcomes:

- Understand how an ultrasonic sensor works.
- Learn to interface a servo motor with an Arduino.
- Develop basic automation skills using sensors and actuators.

This automatic dustbin project helps develop practical knowledge of sensors and motor control, which can be applied in various automated systems.